

# ENT260 – Advanced Solid Modeling for Mechatronics

**Anne Arundel Community College – Spring 2025**

**Instructor:** Prof. Zac Cohen

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**Meeting Time:** Thursday 5:30 PM – 9:40 PM (CALT 354)

**Credits:** 3

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## Course Description:

This course provides an immersive, project-driven approach to mastering SolidWorks for students in the Mechatronics program. Through weekly design challenges, students develop advanced 3D modeling, assembly, and documentation skills aligned with real-world engineering problems. The course also prepares students to pass the **Certified SolidWorks Associate (CSWA)** exam by embedding exam-relevant skills in every challenge. A final team-based design project simulates a professional engineering workflow, integrating mechanical design, manufacturing, and electronics.

By the end of the course, students will be able to:

- Model, assemble, and analyze complex mechanical systems
  - Apply design thinking and design-for-manufacturing (DFM/DFA) principles
  - Simulate motion, perform basic analysis, and generate complete drawing packages
  - Collaborate on a full-system design project relevant to mechatronics
  - Pass the CSWA certification exam
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# Course Outline & Weekly Challenges

## Module 1: Foundations & Design Thinking

### Week 1: Ergonomic Redesign Challenge

#### **Scenario – “The Fatigue Factor”**

A regional tool manufacturer has received a wave of customer feedback: users of its multimeter probes and hand tools report wrist strain and discomfort after prolonged use in tight or awkward spaces—especially among technicians working in automotive repair and HVAC service. The company realizes the original design was based on average hand dimensions from the 1980s and did not consider diversity in grip styles, glove use, or extended use conditions.

You are hired as a junior product designer to lead the ergonomic redesign of one of their common handheld tools. Your challenge is to explore and present several improved design directions that enhance comfort and usability without changing the working interface or tool function. The company wants to see how you apply modern CAD to drive human-centered innovation.

- **Deliverables:**
  - Three concept models showing different ergonomic strategies
  - One-page design rationale explaining your choices
  - Basic dimensioned drawing of the final design
- **Skills Practiced:** Sketching, constraints, extrudes, revolves, fillets, annotation

### Week 2: Parametric Design Challenge – Universal Motor Mount

#### **Scenario – “The Startup That Thinks Big”**

You’ve been hired as a contract CAD designer for a growing robotics startup. They’re launching an educational robotics kit meant for high schools and universities, but they’re facing a logistics headache: every time they change motor types (NEMA 17, 23, or 34), the mounting system must be completely reworked—costing them time, money, and manufacturing errors.

Your job is to create a universal motor mount—a single SolidWorks model that can adapt to all three NEMA sizes using parameters and configurations. It needs to fit a standard 80/20 aluminum extrusion frame and must consider clearances for wiring, airflow, and thermal expansion.

- **Deliverables:**
  - SolidWorks model with parameter-driven configurations
  - Table of design equations and variable definitions
  - Assemblies showing compatibility with provided motor models
  - Explanation of your parameter and DFM strategy
- **Skills Practiced:** Global variables, configurations, equations, parametric design

## Week 3: Feature-Based Design Challenge – Modular Sensor Housing

### Scenario – “Sensors in the Wild”

A sustainability-focused engineering firm is preparing to deploy dozens of environmental sensors throughout a coastal region to monitor temperature, humidity, and light exposure. These sensors need to operate reliably in unpredictable weather conditions.

Your task is to design a modular housing system that allows different sensor modules to plug into a common enclosure, ensuring IP54 protection and flexible mounting options.

- **Deliverables:**
  - Base housing model and two sensor module designs
  - Exploded view and multi-configuration assembly
  - Design intent documentation with sealing and assembly strategies
- **Skills Practiced:** Lofts, sweeps, shelling, feature patterns, mate strategies

## Module 2: Mechanical Systems & Assemblies

### Week 4: Articulating Mount Challenge

#### Scenario – “Vision Under Vibration”

An industrial automation company needs a rugged articulating mount for small machine vision cameras on a factory floor. It must hold a 500g load steadily through vibrations and temperature changes, while allowing easy manual repositioning between tasks. Your design must prioritize stability, durability, and ease of adjustment.

- **Deliverables:**
  - Fully modeled 3DOF arm assembly

- Motion study with joint articulation paths
- Strength analysis of critical parts (manual or simplified FEA)
- 30-second animated demonstration
- **Skills Practiced:** Mates, assemblies, motion study, exploded views, structural design

## Week 5: DFM/DFA Optimization Challenge

### Scenario – "Manufacturing on a Budget"

You are redesigning a motor controller enclosure plagued by high production costs. The current model has undercuts, too many fasteners, and overly complex shapes.

Your challenge is to simplify the design for both 3D printing and injection molding, cutting part count and tooling complexity.

- **Deliverables:**
  - Original vs. optimized model comparison
  - Updated model with snap fits and reduced part count
  - Draft analysis and design for tooling document
  - Brief cost-saving rationale
- **Skills Practiced:** DFM, part simplification, fastener reduction, manufacturability review

## Week 6: Engineering Drawing & Documentation Challenge

### Scenario – "Ready for Production"

A fabrication shop needs full professional documentation for the linear actuator your team has designed. They require detailed dimensioned drawings, tolerances, BOMs, and exploded views to proceed to manufacturing.

Clear, complete, and readable drawings will be essential to your team's success.

- **Deliverables:**
  - Fully dimensioned part drawings
  - BOM and ballooned assembly drawing
  - Exploded view with labeled assembly steps

- Simplified user-facing diagram
  - **Skills Practiced:** Orthographic views, GD&T, BOM creation, exploded/section views
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## Module 3: Applied Mechatronics Design

### Week 7: Electronics Integration Challenge – IoT Enclosure

#### Scenario – "Smart Factory Monitoring"

You are designing a rugged IoT monitoring device housing a Raspberry Pi 4, intended for industrial factories. It must include passive cooling, secure cable routing, sensor mounting ports, and IP54 protection—balancing heat management against sealing.

- **Deliverables:**
  - Solid model with ports, sensor interfaces, and vents
  - Basic thermal simulation or airflow strategy
  - Cable routing diagram with strain relief
  - Maintenance plan for internal components
- **Skills Practiced:** Enclosure design, surface modeling, thermal planning, layout

### Week 8: Precision Mechanism Challenge – Linear Indexer

#### Scenario – "Precision in Motion"

An automated lab system requires a compact, high-precision linear indexing mechanism to move micro-sample trays every 10mm with  $\pm 0.1$ mm repeatability. You are designing a manually or semi-automatically actuated system that locks into place without high friction losses.

- **Deliverables:**
  - Mechanism model and full assembly
  - Motion simulation showing detents or locks
  - Force calculation for actuation effort
  - Component drawings with tolerance notes

- **Skills Practiced:** Precision motion design, mate control, motion studies, force estimation

## Week 9: Additive Manufacturing Challenge – Robotic End Effector

### Scenario – "Printing for Performance"

A research team is building a collaborative robot and needs a 3D-printed end effector optimized for strength and lightweight structure. The final design must be printable with minimal supports on an FDM printer and should have separate versions prepared for SLA printing.

- **Deliverables:**
    - FDM and SLA optimized model variants
    - STL files with slicing screenshots
    - Design for additive manufacturing documentation
    - Performance comparison of print-ready designs
  - **Skills Practiced:** Print orientation, lightweight design, material planning, slicer optimization
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## Module 4: Collaborative Design Project

### Week 10: Capstone Kickoff – Concept Development

#### Scenario – "The Sorting Challenge Begins"

A local distribution center needs a compact, automated system for sorting small electronic components by type and size. Your team's role this week is to brainstorm concept ideas, define system requirements, and create preliminary CAD layouts.

- **Deliverables:**
  - Client-style requirement document
  - Three CAD-aided concept designs
  - Design decision matrix
  - Team structure, task list, and timeline
- **Skills Practiced:** Concept development, team planning, CAD prototyping

## Week 11: Capstone Detailed Design

### Scenario – "From Idea to Integration"

Based on your chosen concept, you will fully develop all mechanical subsystems—including feeders, sorters, and control enclosures—ensuring integration between moving parts and electronics mounting. Motion simulation and interference checking are required.

- **Deliverables:**
  - Fully detailed subsystem models
  - Final integrated assembly
  - Motion and assembly simulations
  - BOM and interface resolution document
- **Skills Practiced:** Detailed modeling, system integration, validation

## Week 12: Capstone Refinement & Documentation

### Scenario – "Handoff to the Client"

You will finalize all deliverables, including manufacturing drawings, maintenance documentation, exploded views, and a polished system demo presentation. Your project package must be ready to hand off to an engineering manager for production.

- **Deliverables:**
    - Finalized CAD and drawings
    - User and assembly manuals
    - Slide deck, demo video, and final reflection
  - **Skills Practiced:** Documentation, communication, presentation preparation
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## Module 5: CSWA Certification & Final Presentation

### Week 13: CSWA Intensive Prep

#### Scenario – "Certification Sprint"

The CSWA exam is approaching! This week you will complete timed SolidWorks part modeling challenges under certification conditions, focusing on model speed, accuracy, and clean feature structure.

- **Deliverables:**
  - Three 30-minute timed part modeling problems
  - Strategy reflection and problem-solving documentation
  - Self-evaluation versus provided solutions
- **Skills Practiced:** Certification alignment, efficient modeling, time management

## Week 14: Final Presentation & CSWA Exam

### Scenario – "Showcase and Certify"

Present your team's final design project to a panel of engineers and industry guests. Demonstrate your system, explain your engineering decisions, and defend your work professionally.

After the presentations, complete your CSWA exam to certify your SolidWorks skills.

- **Deliverables:**
    - 15-minute professional presentation
    - Portfolio submission with all project files
    - CSWA certification completion or exam reflection
  - **Skills Practiced:** Design communication, exam performance, professional presentation
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## Grading Breakdown:

### Assignment Point Values & Assessment Focus

- **Week 1 – Ergonomic Redesign (5%):** Assessed on creativity, completeness of concept models, adherence to ergonomic principles, and quality of annotations.
- **Week 2 – Parametric Motor Mount (5%):** Assessed on parameter logic, model adaptability, and configuration quality.



- **Week 3 – Modular Sensor Housing (5%):** Assessed on IP54 consideration, modularity, and proper use of advanced features.
- **Week 4 – Articulating Mount (5%):** Assessed on functional articulation, joint detail, and simulation clarity.
- **Week 5 – DFM/DFA Optimization (5%):** Assessed on reduction of part complexity, use of manufacturing features, and cost-savings rationale.
- **Week 6 – Drawing Documentation (5%):** Assessed on clarity, accuracy, and completeness of part and assembly drawings.
- **Week 7 – IoT Enclosure Design (5%):** Assessed on thermal strategy, component integration, and enclosure detailing.
- **Week 8 – Linear Indexing Mechanism (5%):** Assessed on precision, motion study execution, and functional layout.
- **Week 9 – Additive End Effector (5%):** Assessed on optimization for print, STL prep quality, and FDM vs SLA design rationale.
- **Weeks 10–12 – Capstone Project (25%):** Assessed on concept development, modeling detail, simulation, documentation, presentation quality, and team collaboration.
- **Week 13 – CSWA Prep (10%):** Assessed on accuracy and efficiency of timed problems, self-assessment, and improvement plan.
- **Week 14 – CSWA Exam & Final Presentation (10%):** Assessed on presentation clarity, team design communication, and CSWA certification or documented exam effort.

| Component | % of Grade | Notes |

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| **Weekly Design Challenges (9)** | **45%** | Practical, CSWA-aligned projects |

| **Capstone Project (Weeks 10–12)** | **25%** | Team design, documentation, and presentation |

| **CSWA Preparation & Certification** | **20%** | Timed modeling, strategy, certification |

| **Participation & Peer Reviews** | **10%** | Feedback, engagement, collaboration |

| **Total** | **100%** |

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## Software & Tools Used:

- ✓ **SolidWorks 2024 (Main Software)**
- ✓ **Motion Study, FEA, Draft Analysis, Thermal Simulation Tools**
- ✓ **Cura, PrusaSlicer, BambuStudio for 3D printing prep**
- ✓ **Collaboration and planning via Canvas & Google Workspace**